Continents' Response to a Global Pandemic

A DS 4002 Case Study by Blake Richardson

In January 2020, the Coronavirus gained international attention as it spread across the entire world. Declared a global pandemic on March 11, 2020, COVID-19 fundamentally changed the way the world handled public health crises, creating an urgent need for effective intervention strategies [1]. Each part of the world faced different challenges—from population density to healthcare capacity— and, therefore, took vastly different approaches in combatting rapidly growing infection and death rates. Consequently, COVID-19's impact varied significantly across these different regions, with mortality rates as high as 4.57% in North America and as low as 2% in parts of Asia and Oceania [2].

In recent months, there have been growing outbreaks of a new infectious disease strain. Global health officials want to review data from previous epidemics to understand which continent's strategies and conditions were most effective in reducing or limiting infections and deaths.

You are a data scientist who has been tasked with reviewing COVID-19 pandemic data. To identify which continent performed best, you will use time series modeling (SARIMA) to analyze each continent's trajectory of deaths and infection cases. You will select a reasonable cutoff date, using data prior to that date to train your model. Then, produce data visualizations to compare these predictions to each continent's actual outcomes, evaluating your model's accuracy. Finally, use your inferential statistic results to determine whether differences in pandemic outcomes across continents are statistically significant and can, therefore, be partly explained by strategic interventions.

[1] F. P. Polack et al., "Safety and efficacy of the BNT162b2 mRNA COVID-19 vaccine," New England Journal of Medicine, vol. 383, no. 27, pp. 2603–2615, Dec. 2020, doi: 10.1056/NEJMoa2034577.

[2] M. N. Zahid and S. Perna, "Continent-Wide Analysis of COVID 19: Total Cases, Deaths, Tests, Socio-Economic, and Morbidity Factors Associated to the Mortality Rate, and Forecasting Analysis in 2020–2021," International Journal of Environmental Research and Public Health, vol. 18, no. 10, p. 5350, May 2021, doi: https://doi.org/10.3390/ijerph18105350.